

## **Khalifa University**

Department of Computer Science

COSC 336 – Introduction to Software Engineering (Fall 2026)

Lab-based Running Project

### **Lab Phase 1**

Initial Plan and Requirement Gathering Document

#### **Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System**

| Field                | Value                                                                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title of the project | Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System                                                                              |
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| GitHub repository    | <a href="https://github.com/duqins/ITS">https://github.com/duqins/ITS</a>                                                                                                     |

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# 1. Introduction

## 1.1 Project Concept

Universities regularly purchase furniture, computers, electronic devices, laboratory equipment, books and office materials while similar usable items sit unused, stored or discarded in other departments. The problem is rarely a shortage of reusable resources; it is the absence of an integrated system that can identify, classify, match, transfer, maintain and track those resources throughout their life cycle.

This project designs and prototypes an internal campus platform, the Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System, that combines a classic, rule-based asset-management core (registration, marketplace, requests, approvals, transfers, maintenance, role-based access, reports) with AI-enhanced services (semantic matching, automatic classification, sustainable-action recommendations, an LLM assistant and generative reporting). The system is built in eight phases over the Fall 2026 semester and gives the team hands-on experience of the full software development life cycle while comparing traditional and AI-augmented engineering approaches.

## 1.2 Document Purpose

This document is the Phase 1 deliverable. It establishes the foundation for all later phases by:

- defining the project problem, scope, objectives, assumptions and expected benefits;
- identifying all stakeholders and mapping them to system features, data needs, permissions, responsibilities and approval decisions;
- describing the current (as-is) process for surplus identification, resource requests, transfer approval, maintenance recording and disposal;
- describing the requirements-gathering methodology, summarising and prioritising the gathered requirements, and giving an initial requirements specification;
- defining measurable success criteria; and
- presenting a preliminary project plan with timeline, milestones, methodology, team structure, task assignments, dependencies, risks and resource allocation.

The document will be refined by the Feasibility Study (Phase 2) and superseded in detail by the Requirements Document (Phase 3). It is written for the lab instructors (client), the development team and future readers of the project repository.

# 2. Project Overview

## 2.1 High-Level Description of the Proposed System

The system is a web-based internal marketplace and asset-life-cycle platform for a university campus. It operates in two layers:

- **Classic baseline layer.** Departments register assets with full metadata, publish surplus or underutilised items, and submit resource requests. Structured search and filtering, reservations, multi-stage approval workflows, transfer and custody tracking,

inspection/maintenance/repair records, end-of-life actions (donation, recycling, disposal), role-based access control and conventional operational reports.

- **AI-enhanced layer.** An asset-classification service suggests categories, tags and standardised descriptions; a semantic matching engine ranks assets against requests using meaning, compatibility, condition, quantity, location, urgency and cost; a sustainable-action recommender proposes retain/transfer/repair/refurbish/donate/disassemble/recycle/dispose with an explanation and confidence; an LLM assistant provides natural-language search, request submission and guidance; and generative reporting turns operational data into monthly summaries and management bulletins.

Every AI output is a recommendation. Authorised human users remain accountable for approvals and high-impact actions (transfer, disposal, donation), and the system must be able to explain each recommendation and fall back to classic behaviour when an AI service is unavailable or low-confidence.

## 2.2 Background and Context

Traditional university asset-management systems record ownership, location, purchase information and disposal status. They do not support intelligent internal reuse, semantic matching between available assets and departmental requests, prediction of future demand, or measurement of the environmental and financial benefits of circular resource management. As a result, departments over-purchase, storage areas fill with idle equipment, and the institution cannot report on avoided purchases, reuse rates or carbon impact. Circular-economy practice and campus sustainability targets create a clear demand for a system that closes this gap.

The project is a lab-based running project for COSC 336 at Khalifa University. It simulates real-world engineering practice: planning, feasibility, requirements, architecture, design, implementation, testing and documentation, with all work tracked in a shared GitHub repository.

## 2.3 Business Goals and Objectives

| #  | Goal                         | Objective (what the system must achieve)                                                                                            |
|----|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| G1 | Reduce unnecessary purchases | Make existing surplus assets visible and easy to request so departments reuse before they buy.                                      |
| G2 | Extend asset life            | Track inspections, maintenance and repairs so assets are kept in service longer and retired only as a last option.                  |
| G3 | Improve matching quality     | Use semantic matching and ranking so requests are met by functionally suitable assets even when terminology differs.                |
| G4 | Ensure accountable workflows | Provide transparent, auditable approval, transfer and custody processes with role-based access.                                     |
| G5 | Measure sustainability value | Calculate and report waste diversion, reuse rate, asset-life extension, financial savings and estimated carbon-emission reductions. |
| G6 | Integrate AI responsibly     | Deliver explainable, confidence-scored recommendations with human oversight and fallback behaviour.                                 |
| G7 | Learning objective           | Apply and compare classic and AI-enhanced software engineering across the full SDLC.                                                |

## 2.4 Project Constraints

| Constraint        | Description                                                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timeline          | Eight phases across the Fall 2026 semester (Phase 1 week of 14 Sep 2026; final submission and demonstration week of 23 Nov 2026). Deadlines are fixed by the course; scope must fit them.                                     |
| Budget            | No monetary budget. Only free/open-source frameworks, free-tier cloud or LLM APIs, and university-provided lab resources may be used.                                                                                         |
| Team resources    | Student team of three members working part-time alongside other courses; limited AI/ML experience; no dedicated designer, tester or DevOps engineer.                                                                          |
| Data              | No access to real university asset data. The prototype will use synthetic and anonymised sample data; sustainability figures (e.g., carbon factors) will use published estimates and must be reported with their uncertainty. |
| Process           | Git and GitHub are mandatory; commit history is graded (10 % of each phase). Regular, meaningful, individually-authored commits are required.                                                                                 |
| Technology        | Web application; must run on a grader's machine from the repository with documented setup. AI features depend on external LLM/embedding APIs with rate limits.                                                                |
| Policy and ethics | Must respect university procurement, ownership, privacy and safety policies; AI must not make final decisions on disposal, donation or transfer.                                                                              |

## 2.5 Scope of the Project

### In scope

- User management and authentication with role-based access for the nine defined roles (Section 3).
- Department management, asset registration and inventory with photographs and supporting documents.
- Circular resource marketplace: publication of surplus/underutilised/unwanted assets visible to authorised campus users.
- Departmental resource requests with category, purpose, specifications, quantity, preferred condition, urgency, location and required date.
- Structured search and filtering; reservations and request tracking.
- Approval workflows for release, reservation, transfer, receipt, donation, recycling and disposal.
- Transfer management (approval, collection/delivery, custody change, new location, confirmation, documents).
- Asset life-cycle history from acquisition to retirement; maintenance and repair module with costs and condition updates.
- Conventional operational reports; notifications; audit logging.

- AI: classification/tagging, semantic matching with compatibility score and explanation, prioritisation/ranking, sustainable-action recommendation, LLM assistant, generative reports.
- Sustainability indicators and dashboard summaries.
- Full documentation across all eight phases and a working prototype with demonstration.

### Out of scope

- Integration with the university's live ERP, finance or procurement systems (interfaces will be simulated).
- Financial transactions, purchasing or payment processing.
- Physical logistics (vehicle scheduling, barcode/RFID hardware), beyond recording collection/delivery arrangements.
- Training custom machine-learning models from scratch; the project uses pre-trained models and hosted LLM/embedding services.
- Mobile native applications (the web UI will be responsive instead).
- Production deployment, high-availability hosting and formal security certification.

## 2.6 Assumptions

- The lab instructors act as the client; their project description and slides are the authoritative statement of needs.
- Synthetic data of realistic size (hundreds of assets, tens of departments and requests) is sufficient to demonstrate all functionality.
- A free-tier hosted LLM and embedding API will remain available throughout the semester, and the team may fall back to a smaller open-source model if not.
- All team members have GitHub accounts, Git installed and access to the shared repository by 15 September 2026.
- Deadlines follow the Phase 1 lab slides (week of 14 Sep 2026 onward); the project description lists a tentative later schedule and the slide dates take precedence unless the instructors announce otherwise.

## 2.7 Expected Benefits

| Beneficiary                            | Benefit                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Departments                            | Faster access to needed equipment without procurement lead time; reduced storage of idle items.              |
| University finance                     | Avoided purchases and quantified savings; better visibility of asset value and utilisation.                  |
| Sustainability office                  | Measured waste diversion, reuse rate and carbon-emission reductions to support campus targets and reporting. |
| Asset custodians and maintenance staff | Single, auditable life-cycle record; structured inspection and repair workflow.                              |
| Administrators                         | Transparent approval trail; role-based control of high-impact actions.                                       |

| Beneficiary  | Benefit                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------|
| Project team | Practical experience of the full SDLC, responsible AI integration and team-based Git workflows. |

### 3. Stakeholder Analysis

#### 3.1 Identification of Stakeholders

Stakeholders were identified from the project description (nine system roles), the lab slides and a walkthrough of the as-is process (Section 4). They fall into four groups: end users, decision makers, the client/oversight group, and the development team.

| Group              | Stakeholder                                   | Interest in the system                                                                                  |
|--------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| End users          | Department representatives                    | Publish surplus assets and request resources on behalf of their department.                             |
| End users          | Requesters (staff, faculty, lab coordinators) | Find, reserve and request assets that meet a specific need.                                             |
| End users          | Asset custodians                              | Maintain asset records, condition and location; release and receive assets.                             |
| End users          | Maintenance staff                             | Receive defect reports, record inspections, repairs and costs, update condition.                        |
| Decision makers    | Administrators (department/college level)     | Approve releases, reservations and transfers; resolve conflicts.                                        |
| Decision makers    | Procurement officers                          | Confirm reuse before new purchase; validate ownership changes.                                          |
| Decision makers    | Finance officers                              | Track asset value, savings, transfer cost and write-offs.                                               |
| Decision makers    | Sustainability officers                       | Define sustainability metrics; review recommendations for donation/recycling/disposal; consume reports. |
| System             | System administrators                         | Manage accounts, roles, configuration, AI service settings, audit logs and backups.                     |
| Client / oversight | Lab instructors (client, sponsor)             | Define requirements, evaluate deliverables and prototype.                                               |
| Client / oversight | University management (indirect)              | Receive management bulletins and sustainability updates.                                                |
| Development        | Project team                                  | Design, build, test, document and demonstrate the system.                                               |

#### 3.2 Stakeholder Roles and Responsibilities

| Stakeholder               | Responsibilities in the process                                                                                                                     | Permissions in the system                                                                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Department representative | Register department assets; flag surplus; submit and track requests; confirm receipt of transferred assets.                                         | Create/edit own department's assets and requests; publish listings; view marketplace; view own department reports. |
| Requester                 | Search marketplace; reserve or request assets; use the assistant; provide feedback on match usefulness.                                             | Read marketplace; create reservations/requests; view own request status.                                           |
| Asset custodian           | Keep asset record accurate; perform hand-over and hand-back; record location and custody changes.                                                   | Edit assigned assets' condition, location, photos, documents; confirm release and receipt.                         |
| Maintenance staff         | Inspect, repair, refurbish; record actions, parts and costs; update condition after service.                                                        | Create inspection/maintenance/repair records for any asset; change condition status.                               |
| Administrator             | Approve or reject release, reservation, transfer, donation, recycling, disposal within their scope; override AI recommendations with justification. | Approve/reject workflow steps; view department-wide data; view recommendation explanations.                        |
| Procurement officer       | Check marketplace before approving new purchases; validate asset acquisition data.                                                                  | Read all listings and requests; annotate requests as 'purchase avoided'; view savings reports.                     |
| Finance officer           | Maintain estimated value, depreciation and transfer cost; approve write-offs.                                                                       | Edit financial fields; approve disposal from a financial perspective; view financial reports.                      |
| Sustainability officer    | Define carbon/waste factors; review end-of-life recommendations; publish sustainability reports.                                                    | Configure sustainability parameters; approve donation/recycling; generate sustainability reports.                  |
| System administrator      | Account provisioning, role assignment, configuration, AI model version tracking, backup, audit.                                                     | Full administrative access; cannot approve business workflows unless also holding a business role.                 |

### 3.3 Stakeholder Needs and Expectations

| Stakeholder               | Needs                                                                 | Expectations of the system                                                                                           |
|---------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Department representative | Quick publication of surplus; simple request form; status visibility. | Publishing an asset takes minutes; AI suggests category and description; clear notifications on every status change. |
| Requester                 | Find functionally suitable items even with different terminology.     | Semantic search returns relevant items; ranked matches with an explanation; can ask the assistant in plain language. |
| Asset custodian           | Accurate, single record per asset; minimal duplicate data entry.      | Life-cycle history is complete and auditable; custody hand-over is confirmed by both parties.                        |
| Maintenance staff         | Structured defect reports and repair logs with cost capture.          | Defect-to-repair workflow with status tracking and condition update.                                                 |

| Stakeholder            | Needs                                                    | Expectations of the system                                                                                         |
|------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Administrator          | Clear approval queue and decision context.               | Approval screen shows request, asset, match score, explanation and sustainability impact; decisions are logged.    |
| Procurement / Finance  | Evidence of avoided purchases and savings.               | Reports quantify avoided purchases, transfer cost and financial savings with traceable calculations.               |
| Sustainability officer | Reliable, explainable sustainability indicators.         | Waste diversion, reuse rate, asset-life extension and carbon estimates with stated assumptions and uncertainty.    |
| System administrator   | Secure, maintainable configuration.                      | Role management, audit logs, AI service configuration and fallback settings.                                       |
| Lab instructors        | Evidence of SDLC practice and responsible AI reflection. | Professional documentation, regular commits by every member, working prototype, explicit classic-vs-AI comparison. |

### 3.4 Stakeholder-to-Feature Mapping

The table maps each stakeholder to the features they use, the data they need, and the approval decisions they own. This mapping drives the role-based access design in later phases.

| Feature                           | Primary users                   | Data needs                                                                                                                           | Approval decision owner                                              |
|-----------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Asset registration and inventory  | Dept. representative, custodian | Name, description, category, quantity, owner, department, location, purchase date, value, condition, availability, photos, documents | — (custodian confirms record)                                        |
| Surplus publication (marketplace) | Dept. representative            | Asset record, reason for surplus, availability window                                                                                | Administrator approves release                                       |
| Resource request                  | Requester, dept. representative | Category, purpose, specifications, quantity, preferred condition, urgency, location, required date                                   | Administrator (requesting side) validates need                       |
| Search and filtering              | All users                       | Category, condition, department, location, quantity, availability, age, value, life-cycle status                                     | —                                                                    |
| AI matching and ranking           | Requester, administrator        | Listings, requests, compatibility score, explanation, transfer/repair cost, sustainability benefit                                   | Administrator accepts or overrides match                             |
| Reservation                       | Requester                       | Asset, requester, period, request link                                                                                               | Custodian / administrator confirms                                   |
| Transfer management               | Custodian, administrator        | Approval, collection/delivery arrangements, custody change, new location, confirmation, documents                                    | Releasing and receiving administrators; custodians confirm hand-over |

| Feature                           | Primary users                                        | Data needs                                                                          | Approval decision owner                                                      |
|-----------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Maintenance and repair            | Maintenance staff, custodian                         | Defect report, inspection, repair actions, parts, cost, condition after service     | Maintenance lead approves repair spend above threshold (TBD)                 |
| Sustainable-action recommendation | Sustainability officer, administrator                | Asset condition, age, repair cost, demand, waste-diversion benefit, carbon estimate | Sustainability officer + finance officer approve donation/recycling/disposal |
| LLM assistant                     | All users                                            | Read access to listings, requests, status, policies scoped by role                  | — (assistant cannot approve anything)                                        |
| Reports (classic and generative)  | Administrators, finance, procurement, sustainability | Aggregated operational and sustainability data                                      | —                                                                            |
| User and access management        | System administrator                                 | Accounts, roles, departments, audit logs, AI configuration                          | System administrator                                                         |

## 4. Study of the Current (As-Is) Process

Because the team has no access to Khalifa University's internal systems, the as-is process below was reconstructed from the project description, publicly available university asset-management practice, and a structured walkthrough exercise in the lab. It will be validated through stakeholder interviews in Phase 2 and Phase 3 (Section 5.1).

| Process                       | Current practice (typical)                                                                                                                                               | Pain points the system must address                                                                                         |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Identifying surplus assets    | A department notices idle equipment during moves, audits or storage clean-outs; sometimes recorded in a spreadsheet or emailed to facilities; often not recorded at all. | No campus-wide visibility; items sit in storage; no consistent classification or condition record.                          |
| Requesting resources          | Staff raise a purchase request through procurement; occasionally ask colleagues informally whether spare equipment exists.                                               | Reuse depends on personal networks; procurement cannot check for existing surplus; requests lack structured specifications. |
| Approving transfers           | Ad-hoc email approval between department heads; asset register updated (if at all) by the custodian afterwards.                                                          | No standard workflow, no audit trail, custody often left on the original department, location data becomes stale.           |
| Recording maintenance         | Maintenance requests raised through a facilities ticket or directly with a technician; repair costs tracked separately in finance.                                       | Maintenance history not linked to the asset record; no basis for repair-vs-replace decisions.                               |
| Donation, recycling, disposal | Disposal committee reviews write-off lists periodically; items may be scrapped although another department could use them.                                               | Disposal is the default; no sustainability assessment; no measurement of waste diversion or carbon impact.                  |
| Reporting                     | Fixed asset register exports; manual spreadsheets for audits.                                                                                                            | No reuse, savings or sustainability metrics; no trend analysis or management insight.                                       |

Target (to-be) process: every asset has one record and one custodian; surplus is published to a campus marketplace; requests are matched semantically and ranked; approvals follow a defined workflow with an audit trail; maintenance and end-of-life events are recorded against the asset; AI recommends sustainable actions and generates reports; humans approve all high-impact decisions.

## 5. Requirements Gathering

### 5.1 Methodology

Requirements are gathered iteratively. The following techniques are used in Phase 1 and continued through Phases 2 and 3:

| Technique                        | Purpose                                                                                                                                                        | Participants / sources                                                                          | Timing                     |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------|
| Document analysis                | Extract mandated functionality, constraints and deliverables.                                                                                                  | Project description (Fall 2026), Phase 1 lab slides, SRS template (IEEE-style) in the appendix. | Phase 1 (complete)         |
| Client interview / Q&A           | Clarify ambiguities (e.g., approval thresholds, sustainability factors, data availability).                                                                    | Lab instructors during lab sessions.                                                            | Phases 1–3                 |
| Stakeholder role-play interviews | Elicit needs for each of the nine roles.                                                                                                                       | Team members assigned as role owners; scripted interview guide.                                 | Phase 2                    |
| Survey                           | Gauge how staff/students would search for and request equipment, and trust AI recommendations.                                                                 | Short online questionnaire to peers and faculty (10–20 responses).                              | Phases 2–3                 |
| Process walkthrough / workshop   | Model as-is and to-be workflows; identify states and approval stages.                                                                                          | Whole team; whiteboard session; output used for use-case and state diagrams.                    | Phase 1 (initial), Phase 3 |
| Benchmarking                     | Study existing asset-management and reuse platforms (e.g., university surplus portals, Warp It, open-source asset trackers) for feature ideas and terminology. | Public product documentation.                                                                   | Phase 2                    |
| Prototyping                      | Validate UI and AI explanation formats with mockups.                                                                                                           | Team and instructors.                                                                           | Phases 4–5                 |

### 5.2 Summary of Gathered Requirements

The requirements gathered so far group into six areas:

- **Asset and inventory management:** full asset record with metadata, photos and documents; life-cycle status; department and custodian ownership.
- **Marketplace and requests:** publication of surplus; structured requests; search/filter; reservations; request tracking.

- **Workflow and governance:** approval stages for release, reservation, transfer, receipt, donation, recycling, disposal; custody and location changes; notifications; audit log; role-based permissions for nine roles.
- **Maintenance and life cycle:** defect reporting, inspection, repair, refurbishment, cost capture, condition update; complete event history.
- **AI services:** classification and tagging; semantic matching with compatibility score and human-readable explanation; prioritisation by suitability, urgency, cost and sustainability; sustainable-action recommendation; LLM assistant; generative reports; confidence information, user overrides, model-version tracking and fallback behaviour.
- **Sustainability and reporting:** waste diversion, avoided purchases, asset-life extension, reuse rate, financial savings, estimated carbon reductions; conventional and AI-generated reports and dashboards.

### 5.3 Prioritisation of Requirements

Requirements are prioritised with the MoSCoW method, aligned to the project description's three tiers: essential prototype functions (Must), desirable extensions (Should), and possible future enhancements (Could/Won't for this semester).

| Priority                   | Meaning                                                                  | Requirement groups                                                                                                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Must have                  | Required for the Phase 6 prototype and final demonstration.              | Authentication and RBAC; department and asset registration; publication; requests; search/filter; reservation; approval and transfer workflow; maintenance tracking; AI classification; AI matching with score and explanation; sustainability indicators; notifications; basic operational and AI-generated reports. |
| Should have                | Expected in the final submission if time allows; adds significant value. | Sustainable-action recommendation with confidence; prioritisation model with transfer/repair cost and carbon estimate; LLM assistant for search and request submission; audit-log viewer; dashboard summaries; user overrides with justification.                                                                     |
| Could have                 | Nice to have; implemented only after all Must/Should items are stable.   | Demand prediction; assistant handling of policy guidance and asset-history questions; monthly management bulletins; image-based classification; accessibility refinements beyond baseline.                                                                                                                            |
| Won't have (this semester) | Explicitly excluded (see scope).                                         | ERP/finance integration; payments; RFID/barcode hardware; custom model training; native mobile apps; production hosting.                                                                                                                                                                                              |

## 6. Initial Requirements Specification

This section gives the initial functional and non-functional requirements. Identifiers are stable and will be carried into the Phase 3 Requirements Document, where they will be expanded with use cases, business rules and acceptance criteria using the SRS template provided in the project appendix. Priority uses MoSCoW (M/S/C).

## 6.1 Functional Requirements – Classic Baseline

| ID    | Requirement                                                                                                                                                                                                                                                                                | Pri. |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| FR-01 | The system shall allow the system administrator to create accounts and assign one or more of the nine roles (department representative, asset custodian, requester, administrator, procurement officer, finance officer, maintenance staff, sustainability officer, system administrator). | M    |
| FR-02 | The system shall authenticate users and enforce role-based access control on every function and data item.                                                                                                                                                                                 | M    |
| FR-03 | The system shall allow administrators to create and manage departments and assign users and assets to them.                                                                                                                                                                                | M    |
| FR-04 | The system shall allow authorised users to register an asset with name, description, category, quantity, owner, department, location, purchase date, estimated value, condition, availability, photographs and supporting documents.                                                       | M    |
| FR-05 | The system shall maintain a life-cycle status for each asset (e.g., registered, in use, published, reserved, in transfer, under maintenance, donated, recycled, disposed, retired) and record every status change with user and timestamp.                                                 | M    |
| FR-06 | The system shall allow a department representative to publish an asset as surplus, underutilised or unwanted so that it is visible to authorised campus users.                                                                                                                             | M    |
| FR-07 | The system shall allow users to submit a resource request specifying category, purpose, specifications, quantity, preferred condition, urgency, location and required date, and to track its status.                                                                                       | M    |
| FR-08 | The system shall provide structured search and filtering by category, condition, department, location, quantity, availability, asset age, estimated value and life-cycle status.                                                                                                           | M    |
| FR-09 | The system shall allow a requester to reserve an available asset and shall prevent conflicting reservations.                                                                                                                                                                               | M    |
| FR-10 | The system shall implement approval workflows for asset release, reservation, transfer, receipt, donation, recycling and disposal, with configurable approver roles per step.                                                                                                              | M    |
| FR-11 | The system shall record for each transfer: approval, collection or delivery arrangements, custody change, new location, receiving confirmation and supporting documents.                                                                                                                   | M    |
| FR-12 | The system shall allow authorised users to report defects, request inspection, record inspection, maintenance, repair and refurbishment actions with costs, and update the asset's condition after service.                                                                                | M    |
| FR-13 | The system shall maintain a complete asset life-cycle history (acquisition, allocation, utilisation, inspection, maintenance, repair, transfer, refurbishment, donation, recycling, retirement, disposal).                                                                                 | M    |
| FR-14 | The system shall send notifications (in-app and/or e-mail) to relevant stakeholders on request submission, match found, approval decision, transfer step, maintenance update and end-of-life action.                                                                                       | M    |
| FR-15 | The system shall generate conventional reports on available assets, open requests, completed transfers, maintenance history, ownership changes, approvals and asset status.                                                                                                                | M    |
| FR-16 | The system shall keep an immutable audit log of all create, update, approve, override and delete actions.                                                                                                                                                                                  | S    |

## 6.2 Functional Requirements – AI-Enhanced Extension

| ID    | Requirement                                                                                                                                                                                                                                                                             | Pri.                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| AI-01 | On asset registration the system shall analyse title, description, specifications and (where available) images and documents to recommend a category, subcategory, standardised tags, possible material type and missing metadata; the user may accept, edit or reject each suggestion. | M                    |
| AI-02 | The system shall semantically match available asset listings with open requests based on meaning rather than exact keywords.                                                                                                                                                            | M                    |
| AI-03 | The matching engine shall consider category compatibility, functional purpose, technical specifications, quantity, asset condition, requested condition, physical location, urgency, required date, transfer cost and repair cost.                                                      | M                    |
| AI-04 | For each recommended asset–request pair the system shall produce a compatibility score (0–100) and a human-readable explanation of the main contributing factors.                                                                                                                       | M                    |
| AI-05 | The system shall rank suitable matches by functional suitability, urgency, transfer cost, repair requirements, waste-diversion benefit and estimated carbon-emission reduction.                                                                                                         | S                    |
| AI-06 | The system shall recommend a sustainable action for an asset (retain, transfer, repair, refurbish, donate, disassemble for components, recycle, or dispose as a last option) with a confidence level and rationale.                                                                     | S                    |
| AI-07 | The system shall provide an LLM-powered assistant supporting natural-language asset search, request submission, asset-history and status questions, policy guidance, sustainability questions and navigation of system functions, scoped to the user's permissions.                     | S                    |
| AI-08 | The system shall apply generative AI to produce summarised monthly reports, sustainability updates, management bulletins, high-demand-category summaries, maintenance insights and lists of recommended actions from operational data.                                                  | M (basic) / S (full) |
| AI-09 | All AI outputs shall be recommendations only; approval of transfers, donations, recycling and disposal shall require an authorised human decision, and users shall be able to override any recommendation with a recorded justification.                                                | M                    |
| AI-10 | The system shall display confidence information for AI outputs, record the model/service version used for each recommendation, and fall back to classic rule-based behaviour when an AI service is unavailable or returns low confidence.                                               | S                    |

### 6.3 Sustainability Measurement Requirements

| ID    | Requirement                                                                                                                                                                                                                            | Pri. |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| SU-01 | The system shall compute and display waste diverted (kg or item count), reuse rate, asset-life extension, number and value of avoided purchases, financial savings and estimated carbon-emission reductions.                           | M    |
| SU-02 | Sustainability calculations shall use configurable factors (e.g., emission factor per category, average replacement cost) maintained by the sustainability officer, and each figure shall state its assumptions and uncertainty range. | S    |
| SU-03 | The system shall present sustainability indicators on a dashboard and include them in generated reports.                                                                                                                               | M    |

### 6.4 Non-Functional Requirements

| ID     | Category                    | Requirement                                                                                                                                                                                                                          | Pri. |
|--------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| NFR-01 | Usability                   | Core tasks (publish asset, submit request, approve transfer) shall be completable by a new user in under 5 minutes with no training; the UI shall follow consistent layouts and provide accessible contrast and keyboard navigation. | M    |
| NFR-02 | Performance                 | Structured search shall return results within 2 s and AI matching within 10 s for a catalogue of at least 1,000 assets and 200 open requests.                                                                                        | S    |
| NFR-03 | Security                    | Passwords hashed; sessions expire; all access checked server-side against role; financial and operational data visible only to authorised roles; protection against common web vulnerabilities (injection, XSS, CSRF).               | M    |
| NFR-04 | Privacy                     | Data minimisation: personal data limited to what is needed for accountability; audit logs and reports avoid exposing personal data beyond role scope.                                                                                | M    |
| NFR-05 | Auditability                | Every approval, override and life-cycle change is traceable to a user, timestamp and (for AI) model version.                                                                                                                         | M    |
| NFR-06 | Explainability              | Every AI recommendation is accompanied by an explanation understandable by a non-technical administrator.                                                                                                                            | M    |
| NFR-07 | Reliability / fallback      | If AI services fail, all classic functions remain available and the UI indicates that AI features are degraded.                                                                                                                      | M    |
| NFR-08 | Maintainability             | Modular, layered architecture with clear separation between UI, application logic, data and AI services; documented APIs; automated tests for core modules.                                                                          | M    |
| NFR-09 | Portability / deployability | The prototype runs from the repository on a clean machine following the README setup in under 15 minutes.                                                                                                                            | M    |
| NFR-10 | Data integrity              | Validation of all inputs; referential integrity between assets, requests, transfers and history; backup/restore procedure documented.                                                                                                | S    |
| NFR-11 | Scalability                 | Design supports growth to campus scale (tens of thousands of assets) without architectural change, even if the prototype is not load-tested at that scale.                                                                           | C    |
| NFR-12 | Fairness and ethics         | Matching and recommendations shall not systematically disadvantage any department; recommendation logic is documented and reviewable.                                                                                                | S    |

## 7. Measurable Project Success Criteria

| Area                   | Criterion                                                                                | Target (prototype)                      | How measured                                            |
|------------------------|------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------|
| Functional performance | All Must-have functional requirements implemented and passing tests                      | 100 % of Must; ≥ 70 % of Should         | Requirements traceability matrix; test report (Phase 7) |
| Functional performance | End-to-end workflow: register → publish → request → match → approve → transfer → receipt | Completes without manual database edits | System test scenario                                    |
| User acceptance        | Task success and satisfaction in user-acceptance testing                                 | ≥ 90 % task completion; mean SUS ≥ 70   | UAT with ≥ 6 participants across roles                  |

| Area                            | Criterion                                                                                           | Target (prototype)                                  | How measured                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------|
| Match usefulness                | Top-3 semantic match contains a suitable asset (judged by tester)                                   | ≥ 80 % of test requests                             | Labelled synthetic request set (≥ 30 requests) |
| Match usefulness                | Explanation rated understandable and relevant                                                       | ≥ 4/5 average rating                                | UAT questionnaire                              |
| Classification quality          | AI-suggested category accepted by user                                                              | ≥ 75 % acceptance                                   | Log of accept/edit/reject on test set          |
| Avoided purchases               | System records avoided purchases with value                                                         | Report produced; calculation traceable              | Report validation against sample data          |
| Reuse / repair / sustainability | Reuse rate, asset-life extension, waste diversion, carbon estimate computed with stated uncertainty | All five indicators on dashboard and in report      | Validation of calculations (Phase 7)           |
| Responsible AI                  | Every high-impact action requires human approval; fallback works with AI disabled                   | 100 % of workflows                                  | Security/workflow tests                        |
| Process                         | Regular individually-authored commits by every member                                               | ≥ 2 meaningful commits/member/week; PR-based merges | GitHub history review                          |
| Documentation                   | All eight phase deliverables submitted on time                                                      | 8/8                                                 | Course submissions                             |

## 8. Initial Project Plan

### 8.1 Proposed Methodology

The team will follow an Agile approach (Scrum-style) constrained by the course's phase gates: each course phase is treated as one or more sprints with a defined deliverable, a sprint planning session in the first lab of the phase, a mid-phase check-in, and a review with the instructors at submission. Agile is preferred over Waterfall because AI features carry technical uncertainty (API availability, match quality) that requires early prototyping and iteration, while the fixed course deadlines and document deliverables still give the overall project a milestone-driven backbone. Requirements and design documents will be versioned in the repository and updated as learning occurs.

### 8.2 Preliminary Timeline and Milestones

| Phase | Due (labs of week of) | Milestone / deliverable                         | Weight | Key tasks                                                                                                                                                |
|-------|-----------------------|-------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | 14 Sep 2026           | Initial Plan and Requirement Gathering Document | 10 %   | Problem, scope, stakeholders, as-is study, initial requirements, plan. Repo created and collaborators added by 15 Sep.                                   |
| 2     | 21 Sep 2026           | Feasibility Document                            | 10 %   | Technical, financial, operational, schedule, data/AI and legal/ethical feasibility; technology stack decision; LLM/embedding API selection and fallback. |

| Phase | Due (labs of week of) | Milestone / deliverable                            | Weight | Key tasks                                                                                                                                                                               |
|-------|-----------------------|----------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | 5 Oct 2026            | Requirements Document                              | 10 %   | Full SRS (template in appendix): actors, states, business rules, FR/NFR, AI and sustainability requirements, use-case diagram and descriptions, prioritisation.                         |
| 4 & 5 | 26 Oct 2026           | Design Document (Architecture and Detailed Design) | 20 %   | Layered architecture, component/deployment diagrams, DFDs, ERD, class/sequence/activity/state diagrams, AI workflow diagrams, interface contracts, mockups, security boundaries.        |
| 6     | 16 Nov 2026           | Draft Implementation with Major Features           | 20 %   | Working prototype: auth/RBAC, departments, assets, marketplace, requests, search, AI classification and matching, approvals, transfers, maintenance, sustainability dashboard, reports. |
| 7     | 23 Nov 2026           | Details of Test Cases                              | 10 %   | Unit, integration, system, AI-quality, sustainability-calculation and user-acceptance tests; limitations of test data reported honestly.                                                |
| 8     | 23 Nov 2026           | Final Project, Presentation and Demonstration      | 20 %   | Final code, configuration, documentation across all phases, presentation and live demo.                                                                                                 |

*Note: the project description lists a tentative schedule starting one week later (Phase 1 in the week of 20 September 2026). This plan follows the Phase 1 lab slides, which are the more recent instruction; the team will confirm the dates with the instructors.*

### 8.3 Team Structure and Roles

Roles rotate partially so that every member gains experience across the SDLC, but each member owns one area for accountability and for clear individual authorship in the commit history.

| Team member     | Primary role                                 | Owns (modules / documents)                                                                                                           | Secondary role                            |
|-----------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Sultan Almheiri | Project lead / Scrum master and backend lead | Planning documents, sprint coordination, repository administration, data model, workflow engine (approvals, transfers), integration  | Sustainability calculations and reporting |
| Zayed Alfadli   | Requirements, UX and QA lead                 | Requirements documents, use cases, mockups, UI implementation, test plan and test cases, UAT, user documentation                     | Frontend developer                        |
| Ghith Alhinaai  | Architecture and AI integration lead         | System architecture, security, classification, semantic matching, recommendation, LLM assistant, generative reporting, AI evaluation | DevOps / documentation                    |

### 8.4 Task Assignments for Phases 1–3

| Task                                                                                                                                   | Owner                                          | Due                      |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------|
| Finalise Phase 1 document and submit                                                                                                   | Project lead (Sultan); all contribute sections | Week of 14 Sep           |
| Create GitHub repository, add instructor and lab engineer as collaborators, add README, .gitignore, licence, branch protection on main | Project lead (Sultan)                          | 15 Sep                   |
| Set up Git identity and first commit for every member                                                                                  | Each member                                    | 15 Sep                   |
| Technology stack proposal and AI API evaluation (free tiers, rate limits, fallback)                                                    | AI/architecture lead (Ghith)                   | Week of 21 Sep           |
| Feasibility document (six feasibility dimensions)                                                                                      | Requirements lead (Zayed) drafts, all review   | Week of 21 Sep           |
| Stakeholder interviews and survey                                                                                                      | Requirements/UX lead (Zayed)                   | Weeks of 21 Sep – 28 Sep |
| Use-case diagram and detailed use-case descriptions                                                                                    | Requirements lead (Zayed)                      | Week of 5 Oct            |
| Full SRS using appendix template                                                                                                       | Requirements lead (Zayed) owns, all contribute | Week of 5 Oct            |
| Synthetic data set design (assets, departments, requests)                                                                              | Project lead (Sultan) + AI lead (Ghith)        | Week of 5 Oct            |

## 8.5 Dependencies

- Phase 2 stack decision depends on confirmation of available LLM/embedding API access (free tier or instructor-provided key).
- Phase 3 SRS depends on interview/survey results and instructor answers on approval thresholds and sustainability factors.
- Phase 4–5 design depends on the finalised data model and the chosen AI service interfaces.
- Phase 6 implementation depends on the synthetic data set and on all members having a working local environment.
- Phase 7 AI-quality testing depends on a labelled test set created during Phase 6.
- All phases depend on continuous, individually-authored commits to satisfy the GitHub grading criterion.

## 8.6 Risks and Mitigation

| #  | Risk                                                | Likelihood | Impact | Mitigation                                                                                                               |
|----|-----------------------------------------------------|------------|--------|--------------------------------------------------------------------------------------------------------------------------|
| R1 | LLM/embedding API unavailable, rate-limited or paid | Medium     | High   | Evaluate two providers in Phase 2; design AI as a pluggable service; keep a small open-source model or keyword fallback. |
| R2 | Scope too large for the semester                    | High       | High   | Strict MoSCoW; Must-have items first; Should/Could only after prototype is stable.                                       |
| R3 | Uneven contribution / commit history flagged        | Medium     | High   | Module ownership per member; PR-based workflow; weekly commit check by project lead.                                     |

| #  | Risk                                              | Likelihood | Impact | Mitigation                                                                                                     |
|----|---------------------------------------------------|------------|--------|----------------------------------------------------------------------------------------------------------------|
| R4 | Match quality poor on synthetic data              | Medium     | Medium | Build a labelled test set early; iterate on prompts/embeddings; report limitations honestly.                   |
| R5 | Sustainability figures unreliable                 | Medium     | Medium | Use published factors; state assumptions and uncertainty ranges; sustainability officer role can edit factors. |
| R6 | Security weaknesses in RBAC                       | Medium     | High   | Server-side authorisation on every endpoint; security test cases in Phase 7.                                   |
| R7 | Team member unavailable (illness, exams)          | Medium     | Medium | Secondary role for every area; documentation in repo so work can be picked up.                                 |
| R8 | Schedule ambiguity between slides and description | Low        | Medium | Confirm with instructors in the first lab; plan to the earlier date.                                           |
| R9 | Secrets (API keys) committed to repository        | Medium     | High   | .gitignore from day one; .env files; rotate key immediately if leaked.                                         |

## 8.7 Resource Allocation

- **People:** Three students, approximately 6–8 hours per member per week, rising to 10+ in Phases 6–8.
- **Tools:** GitHub (repository, issues, pull requests, projects board); VS Code with GitHub Copilot (student plan); draw.io/PlantUML for UML; a web framework and relational database chosen in Phase 2; hosted LLM/embedding API (free tier); LibreOffice/Word for documents.
- **Infrastructure:** local development machines; university lab computers; optional free-tier cloud hosting for the demo.
- **Data:** team-generated synthetic asset, department, request and maintenance data; published sustainability factors.
- **Budget:** AED 0 – all tools on free or student tiers.

## 8.8 Version Control and Collaboration Workflow

- One repository, created on GitHub with README, .gitignore and licence; instructor and lab engineer added as collaborators.
- Branch protection on main: no direct commits; every change via a feature branch and pull request with at least one reviewer.
- Conventional Commits messages (feat, fix, docs, test, refactor, chore); one logical change per commit.
- Each member commits under their own verified GitHub identity so authorship is visible.
- Documents for every phase are written in Markdown in the repository (docs/phaseN/sections) and committed incrementally as they are written; the Word/PDF versions are generated from them for submission.
- Secrets are never committed; API keys live in .env files excluded by .gitignore.

## 9. Classic vs. AI-Enhanced Perspective (Initial Reflection)

The project description requires the team to compare the two perspectives explicitly. This initial position will be revisited in every later phase.

| Aspect               | Classic (rule-based) approach                             | AI-enhanced approach                                                                     | Initial position                                                                                        |
|----------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Search               | Exact keyword/category filters; predictable, transparent. | Semantic search finds functionally similar items with different wording.                 | Provide both; classic filters remain the ground truth, AI adds recall.                                  |
| Matching             | Checks category, quantity, availability.                  | Ranks alternatives by specification, condition, urgency, location, cost, sustainability. | AI ranks; rules enforce hard constraints (availability, permissions).                                   |
| Classification       | User selects from fixed taxonomy.                         | Model suggests category/tags from free text and images.                                  | AI suggests, user confirms; taxonomy stays controlled.                                                  |
| Life-cycle decisions | Policy thresholds (age, repair cost).                     | Recommendation with confidence and explanation.                                          | AI advises; humans decide on disposal, donation, transfer.                                              |
| Reporting            | Fixed totals and tables.                                  | Narrative summaries, anomalies, suggested actions.                                       | Generative report built on top of classic figures so numbers stay verifiable.                           |
| Risks                | Rigid; misses reuse when terminology differs.             | Uncertain, opaque, possibly biased; depends on external services.                        | Explainability, confidence, override, model-version tracking and fallback are Must/Should requirements. |

Decisions that must remain under human control: approving transfers and custody changes; authorising donation, recycling and disposal; approving repair spend; changing financial values; changing user roles. AI must never execute these automatically.

## 10. Open Questions for the Client

- Are the Phase 1 lab-slide dates (from 14 Sep 2026) or the description's tentative dates (from 20 Sep 2026) authoritative?
- Is an LLM/embedding API key provided by the course, or should the team rely on free tiers?
- Are there preferred technologies (language, framework, database) or is the choice free?
- What approval thresholds apply (e.g., asset value above which finance must approve a transfer or repair)?
- Should sustainability factors (carbon per category) follow a specific reference source?

## 11. References

- Khalifa University, Department of Computer Science. COSC 336 Lab-based Running Project – Project Description, Fall 2026 (Project-September2026.pdf).
- COSC 336 Lab Phase 1 slides – Initial Plan and Requirement Gathering Document (COSC336\_Circular\_Campus\_Project\_Fall2026.pdf).

- IEEE-style Software Requirements Specification template (appendix of the project description).
- Sommerville, I. Software Engineering, 10th ed., Pearson.
- Git, GitHub and Copilot Onboarding guide (DOC-VC-01).